

Sequential Training Plan: AI Engineering & BPS (6+1 Months)

Project: Integration of Machine Learning and PIML in Building Performance Simulation

Structure: Sequential (Physical Domain → Artificial Intelligence → Integration)

Total Duration: 7 Months (6 Months Course + 1 Month Capstone)

Pedagogical Strategy: "Domain First"

In this extended model, we adopt the "Domain First" approach:

1. **Phase 1 (Months 1-3):** The student masters the "problem" (Physical Simulation and Python). They experience the challenge of reading massive manuals and manipulating complex files.
2. **Phase 2 (Months 4-6):** The student receives the "tool" (Generative AI). They learn to automate what they learned in the previous phase.
3. **Capstone (Month 7):** Full integration.

PREPARATION (Zero Day)

Status:  **MANDATORY PREREQUISITE.** Must be resolved before Day 1.

1. Access and Accounts

- [] **Google Cloud:** Create a project with a personal @gmail.com account and enable billing (Free Tier).
- [] **Coursera for USP:** Mandatory login via the [Partnership Portal](#).
- [] **Google Developer Program:** Register for the [Developer Profile](#) (Personal Account) to unlock free labs.

2. Local Infrastructure (Workstation)

- [] **Core:** Python 3.10+, EnergyPlus 24.1, VS Code, and [Google Cloud SDK](#).
- [] **Python Libraries:** pip install besos energyplus-api-helpers accim eppy pandas scikit-learn matplotlib.
- [] **VS Code Cloud Code:**
 - Install the "Google Cloud Code" extension.
 - Login on the bottom bar with your @gmail.com account.
 - *Test:* Run gcloud auth print-access-token in the integrated terminal.

PHASE 1: THE PHYSICAL DOMAIN (TRACK B)

Objective: Transform the student into a Building Performance Simulation (BPS) specialist via Python.



MONTH 1: Python for Engineering & EnergyPlus

Focus: Abandon the graphical interface and master data manipulation.

- [] **Core Skill: Python Scientific Stack.** Focus on pandas for time-series manipulation (.epw weather files and outputs).
- [] **Reading:** [EnergyPlus Python API Documentation](#).
- [] **Essential Doc:** [Eppy Documentation](#). Tutorial on modifying IDF fields (geometry, materials) via script.
- [] **Practical Task:** "Hello World" without GUI.
 - Script: run_sim.py. Loads an IDF, runs EnergyPlus, and plots a temperature chart using matplotlib.

Professor Checkpoint: "Show me a script that runs the simulation and tells me the maximum temperature of the year, without opening Excel."



MONTH 2: Automation and Data Generation (Big Data)

Focus: Scale from 1 simulation to 1,000 simulations (The foundation for training IAs).

- [] **Methodology: Stochastic Sampling.** Study *Latin Hypercube Sampling (LHS)*.
- [] **Methodology: Modular JSON–Python Workflows** (Osei-Owusu et al., 2025). [DOI](#).
- [] **Methodology: Surrogate Optimization Pipeline** (Kubwimana & Najafi, 2023). [DOI](#).
 - *Concept:* How to create the synthetic dataset to train the lightweight model.
- [] **Tool:** [BESOS Documentation](#). Read the section on integration with EnergyPlus.
- [] **Case Study:** [ACCIM Library](#). Understand how to inject tropical adaptive comfort logic into models.
- [] **Challenge:** Create a script that generates 50 variations of a building (changing orientation and glass type) and saves the results in a consolidated CSV.

Professor Checkpoint: "I want to see the CSV with the results of the 50 simulations. Do the data look physically coherent?"



MONTH 3: The Theoretical Frontier (PIML)

Focus: Understand the State of the Art before entering AI practice.

- [] **Deep Reading:** *PIML for Building Energy Modeling* (Ma et al., 2024). [DOI](#).
- [] **Deep Reading:** *Physics-constrained graph modeling* (Yang et al., 2024). [DOI](#).
- [] **Modular Innovation:** *Physics-Informed Modularized Neural Networks* (Jiang et al., 2025). [DOI](#).
- [] **Tropical Application:** *Evaluating retrofitting in severe regions* (Wang et al., 2025). [DOI](#).
 - *Context:* Justification for climatic adaptation.
- [] **Application:** *AI-Driven Prediction & Thermal Resilience* (Mehraban et al.). [Link MDPI](#).

Professor Checkpoint: "Explain how PIML differs from a standard Neural

Network. Where does 'Physics' fit into the equation?"

PHASE 2: ARTIFICIAL INTELLIGENCE (TRACK A)

Objective: Introduce the "Brain" (Google Cloud Vertex AI) to manage the "Body" (Simulation) created in Phase 1.

MONTH 4: Generative AI Fundamentals & Prompting

Focus: Learn to "speak" to the machine and understand responsibility.

- [] **Course (Technical):** [Introduction to Generative AI](#).
- [] **Course (Technical):** [Introduction to Large Language Models](#).
- [] **Course (Strategic):** [Generative AI for Everyone](#) (Andrew Ng).
- [] **Lab:** [Prompt Design in Vertex AI](#). Test Temperature=0 for scientific codes.
- [] **Inspiration Resource:** [Anthropic System Prompts](#).
- [] **Ethics & UX Reading:** [Google People + AI Guidebook](#).

Professor Checkpoint: "Use Gemini to fix a bug in your simulation script from Month 2. Show me the prompt you used."

MONTH 5: RAG - The Digital Librarian

Focus: Solve the pain of reading massive manuals (experienced in Month 1).

- [] **Advanced Course:** [Generative AI with Large Language Models](#) (DeepLearning.AI).
- [] **Project:** "Chat with the Manual". Ingestion of the *EnergyPlus Engineering Reference* PDF into **Vertex AI Agent Builder**.
- [] **Concept:** Embeddings and Vector Search. [Vertex AI Docs](#).
- [] **Resource:** [Grounding with Google Search](#). Ensure AI cites the standard page.

Professor Checkpoint: "Ask your Agent for the formula of the Perez sky model. It must respond by citing the manual page."

MONTH 6: Function Calling & Integration (The Bridge)

Focus: Where the magic happens. The AI learning to use the Python tools from Phase 1.

- [] **Technical Path:** [Generative AI for Developers](#) (Google Cloud Learning Path).
- [] **MLOps Specialization:** [Machine Learning on Google Cloud](#).
 - *Focus:* Production pipeline modules only.
- [] **Key Concept: Function Calling.** Teach Gemini it has a `run_energyplus(idf_path)` function.
- [] **Tutorial:** [Vertex AI SDK for Python](#).
- [] **Scientific Validation:** [Vertex AI Evaluation Service](#).
 - *Use:* Measure if the agent is hallucinating in data summaries.
- [] **Prototyping Tool:** [Streamlit](#).

- [] **Framework:** [LlamaIndex Documentation](#).

Professor Checkpoint: "I want to see the AI generating the structured JSON that your Python script can read without error."



PHASE 3: CAPSTONE (FULL INTEGRATION)



MONTH 7: The Autonomous Researcher Agent (FAPESP Phase 4 MVP)

Goal: Develop the "Orchestrator Agent" described in Objective 3.2.4 of the FAPESP project ("Cognitive Automation").

Week 1: Agent Architecture and Design

- [] **Tools Definition:** Map which Python scripts from *Phase 1* will become tools.
 - Ex: `get_weather_data(city)`, `modify_idf_geometry(width, height)`, `run_simulation()`.
- [] **System Prompt (The Persona):** Write instructions defining the Agent as a "Senior Simulation Engineer" following the NBR 15.575 standard.
- [] **State Flow:** Draw the diagram: *Receive Question* → *Plan Simulation* → *Execute (EnergyPlus)* → *Analyze (CSV)* → *Respond*.

Week 2: Tool Implementation (Backend)

- [] **Python Wrapper:** Refactor Month 2 automation scripts to accept strict inputs (JSON) and return clean outputs for the LLM.
- [] **Error Handling:** If EnergyPlus fails, the script must return a clear error message so the Agent can attempt to correct the parameter and retry (Self-Correction).

Week 3: Orchestration (The "Brain")

- [] **Function Calling in Vertex AI:** Connect Gemini Pro to the created Python tools.
- [] **RAG Integration:** Connect the Agent to the *Datastore* (created in Month 5) to consult the NBR 15.575 standard when analyzing results.
 - *Scenario:* "Is the operative temperature of 28°C acceptable according to the standard?" (Agent consults PDF and responds).

Week 4: Validation and Final Demo

- [] **Pilot Experiment:** Run the "Glass Optimization in Manaus" scenario.
 1. **Input:** "I want to optimize the glass for Manaus".
 2. **Autonomous Action:** Agent runs 3 simulations (Single, Double, Low-E Glass).
 3. **Output:** Agent generates a final report comparing air conditioning consumption.
- [] **Deliverable:** A documented Jupyter Notebook or Streamlit App demonstrating the full cycle.

Final Checkpoint: The prototype must demonstrate the **Question -> Simulation**

-> **Answer** cycle without human intervention in the middle of the process.



Reference Library (Active Links)

Resource	Type	Direct Link
Coursera for USP	Access	Partnership Portal
Google Cloud Skills	Labs	Access Labs
EnergyPlus API	Doc	ReadTheDocs
BESOS	Lib	Documentation
Eppy	Lib	Documentation
Streamlit	Lib	Documentation
Ma et al. (2024)	Paper	PIML for BEM
Osei-Owusu (2025)	Paper	Modular Workflows
Jiang et al. (2025)	Paper	Modularized PIML